

LUNAR ICE MISSION PLANNER

Automated Mission Analysis Report

Mission Date	2026-06-27 21:24 UTC
Target Region	Chandrayaan-2 Dataset
Instruments Used	DFSAR (Radar) + OHRC (Imaging)
Mission Status	ANALYSIS COMPLETE

1. ABSTRACT

This report presents the results of an automated lunar ice detection and mission planning analysis conducted using Chandrayaan-2 DFSAR radar data and OHRC high-resolution imagery. The analysis pipeline computes Circular Polarization Ratio (CPR) and Degree of Polarization (DOP) to identify potential water-ice deposits in permanently shadowed regions near the lunar south pole. A safe landing site and optimized rover path to the ice deposits are subsequently computed.

2. ICE DETECTION RESULTS

Parameter	Value	Interpretation
CPR Threshold	1.00	Values > 1.0 indicate ice
DOP Threshold	0.30	Values < 0.3 suggest ice
Ice-Positive Pixels	39,370	Pixels exceeding thresholds
Estimated Ice Area	35.433 km ²	Surface area of ice deposits
Estimated Ice Volume	70,866,000 m ³	Volume at 2m depth assumption
Water Equivalent	64,984,122 tonnes	Potential water resource

3. LANDING SITE ANALYSIS

Parameter	Value	Safety Criterion
Grid Position (Row, Col)	(243, 53)	Best scored location
Site Safety Score	99.7%	Higher is better
Slope at Site	0.0°	Must be < 15°
Hazard Level	0.006	Must be < 0.7
Landing Assessment	APPROVED	Mission go/no-go

4. ROVER PATH PLANNING

Parameter	Value
Path Algorithm	A* (Optimal Pathfinding)
Total Distance	0.000 km
Distance (meters)	0.0 m
Average Slope	0.0°
Path Waypoints	0
Estimated Traverse Time	0.0 hours (at 0.1 m/s)

5. MISSION SUCCESS ASSESSMENT

Overall Mission Success Probability: **99.9%**

This score is computed from four weighted components: ice detection confidence (35%), landing site safety (30%), rover path feasibility (20%), and resource value assessment (15%). A score above 70% is considered mission-viable.

6. METHODOLOGY

Step 1 — Radar Processing (DFSAR): The DFSAR compact polarimetry data is read and decomposed into HH and HV polarization channels.

Step 2 — CPR Calculation: $CPR = HV / HH$. Regions with $CPR > 1.0$ are flagged as potential ice bearing, consistent with findings from Spudis et al. (2013) and ISRO's Chandrayaan-1 Mini-RF observations.

Step 3 — DOP Calculation: $DOP = (HH - HV) / (HH + HV)$. Low DOP in permanently shadowed regions corroborates the ice interpretation.

Step 4 — Terrain Analysis (OHRC): Slope and roughness maps are derived from the OHRC imagery gradient to identify safe, flat landing zones.

Step 5 — Landing Site Selection: A weighted scoring model combines ice proximity, slope safety, and hazard avoidance to select the optimal landing coordinate.

Step 6 — Rover Path (A*): The A* algorithm finds the shortest safe path from the landing site to the ice deposit, penalizing high-slope and high-hazard terrain.